

DSA0601 • DATA HANDLING AND VISUALIZATION LAB EXPERIMENT RECORD

Set 14: Survey Response Analysis

Department	AIML
Course Code / Course Name	DSA0601 / Data Handling and Visualization
Experiment	Data Visualization using R and Tableau
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Aim

To analyze survey response data using a stacked bar chart, a radar chart and a tabular representation, and to develop a Tableau dashboard combining the visualizations for interactive exploration of survey results.

Dataset: Survey Responses

Respondent	Question 1	Question 2	Question 3
1	A	B	C
2	B	A	D
3	C	A	B

1. Create a stacked bar chart to visualize the distribution of responses for Question 1. Label the chart elements.
2. Generate a radar chart to show the overall pattern of survey responses across all questions.
3. Build a table to show the survey responses of each respondent. Label the table elements.
4. Develop a Tableau dashboard combining the stacked bar chart, radar chart, and the table for interactive exploration of survey results.

PROGRAM 1: Stacked Bar Chart for Question 1

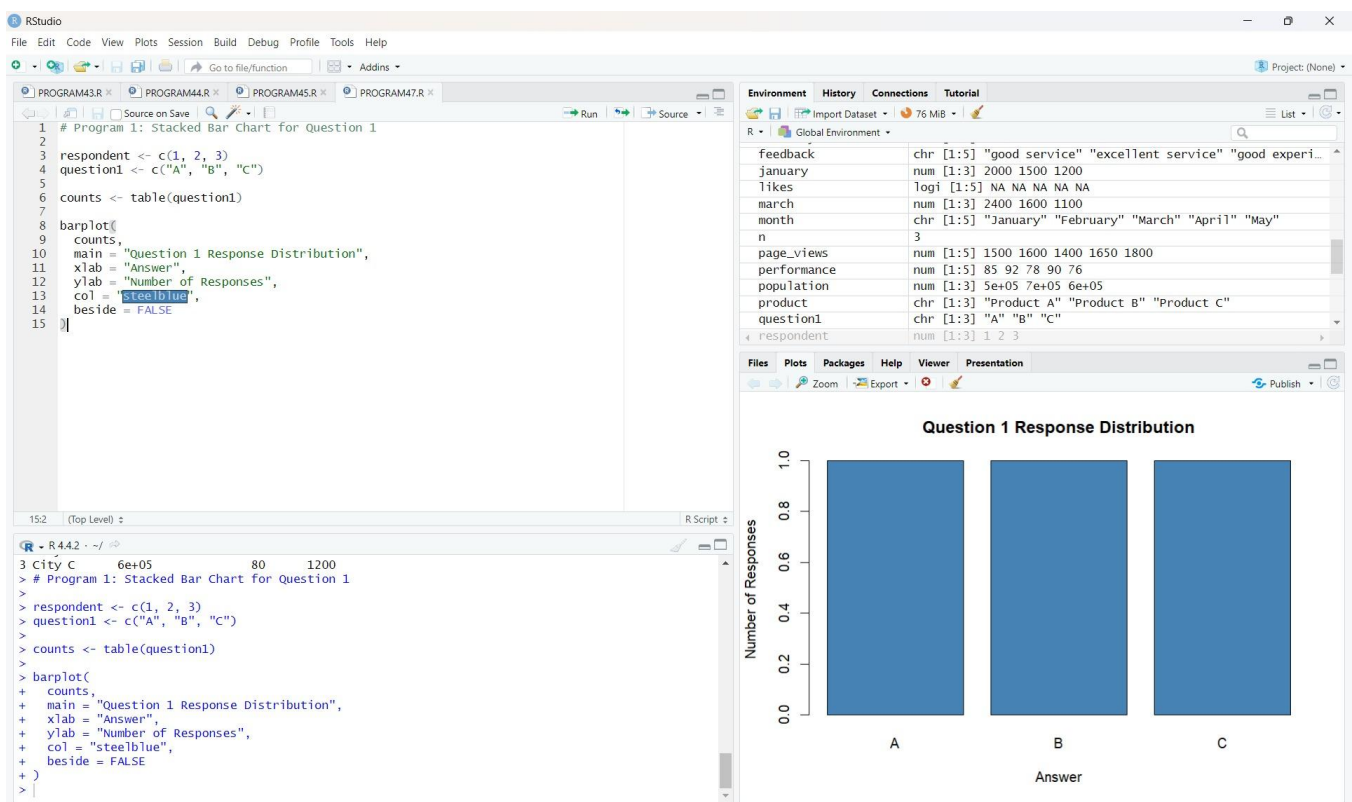
Create the survey responses and visualize the distribution of answers for Question 1. Label the chart elements.

```
# Program 1: Stacked Bar Chart for Question 1
respondent <- c(1, 2, 3)
question1 <- c("A", "B", "C")

counts <- table(question1)

barplot(
  counts,
  main = "Question 1 Response Distribution",
  xlab = "Answer",
  ylab = "Number of Responses",
  col = "steelblue",
  beside = FALSE
)
```

Output 1: Question 1 Response Distribution



Observation: The bar chart shows that answers A, B, and C each received one response for Question 1.

PROGRAM 2: Radar Chart of Survey Responses

Generate a radar chart to show the overall pattern of responses for all survey questions.

```
# Program 2: Radar Chart of Survey Responses
answers <- c("A", "B", "C", "D")
survey_data <- rbind(
  "Question 1" = table(factor(c("A", "B", "C"), levels = answers)),
  "Question 2" = table(factor(c("B", "A", "A"), levels = answers)),
  "Question 3" = table(factor(c("C", "D", "B"), levels = answers))
)

max_value <- max(survey_data)
angles <- seq(0, 2 * pi, length.out = length(answers) + 1)

# Draw radar grid and answer labels
plot(
  0, 0, type = "n", asp = 1, axes = FALSE, xlab = "", ylab = "",
  xlim = c(-(max_value + 0.5), max_value + 0.5),
  ylim = c(-(max_value + 0.5), max_value + 0.5),
  main = "Overall Pattern of Survey Responses"
)
for (r in 1:max_value) {
  lines(r * cos(angles), r * sin(angles), col = "gray")
}
text(
  (max_value + 0.3) * cos(angles[-5]),
  (max_value + 0.3) * sin(angles[-5]),
  labels = colnames(survey_data)
)

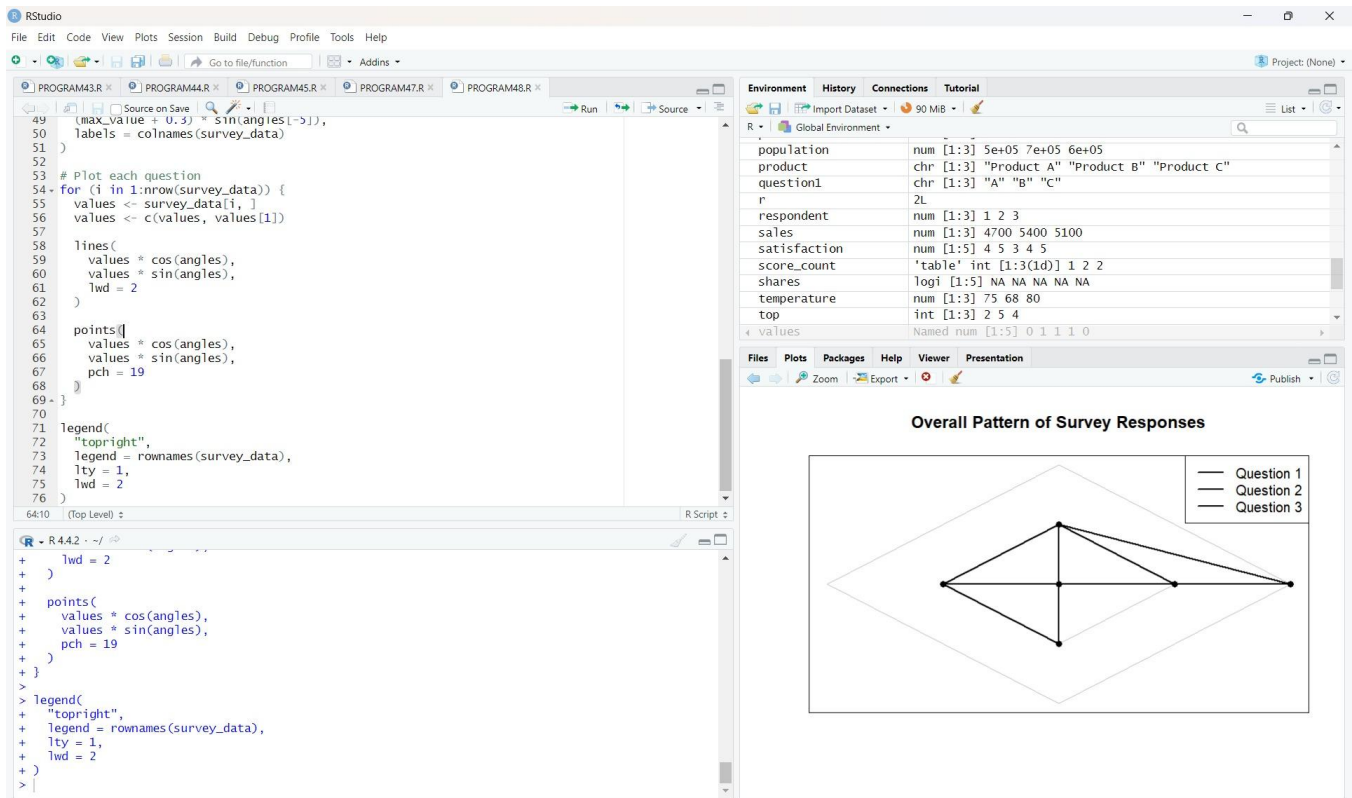
# Plot each question
for (i in 1:nrow(survey_data)) {
  values <- survey_data[i, ]
  values <- c(values, values[1])

  lines(
    values * cos(angles),
    values * sin(angles),
    lwd = 2
  )

  points(
    values * cos(angles),
    values * sin(angles),
    pch = 19
  )
}

legend(
  "topright",
  legend = rownames(survey_data),
  lty = 1,
  lwd = 2
)
```

Output 2: Overall Pattern of Survey Responses



Observation: Question 2 has the highest concentration of responses on a single answer (A = 2), while Questions 1 and 3 are spread across different answers.

PROGRAM 3: Survey Response Table

Build a table to show the survey responses of each respondent. Label the table elements.

```
# Program 3: Survey Response Table
survey_data <- data.frame(
  Respondent = c(1, 2, 3),
  Question_1 = c("A", "B", "C"),
  Question_2 = c("B", "A", "A"),
  Question_3 = c("C", "D", "B")
)

print(survey_data)
```

Output 3: Survey Response Table

The screenshot shows the RStudio interface with the following components:

- Source Editor:** Contains the R script for Program 3, which creates a data frame with 3 respondents and 3 questions.
- Console:** Shows the execution of the script, resulting in a printed table of survey responses.
- Environment Pane:** Lists the variables created: `survey_data` (3 obs. of 4 variables) and `ui` (List of 4).
- Files Pane:** Shows the project structure with files like `PROGRAM43.R`, `PROGRAM44.R`, etc.

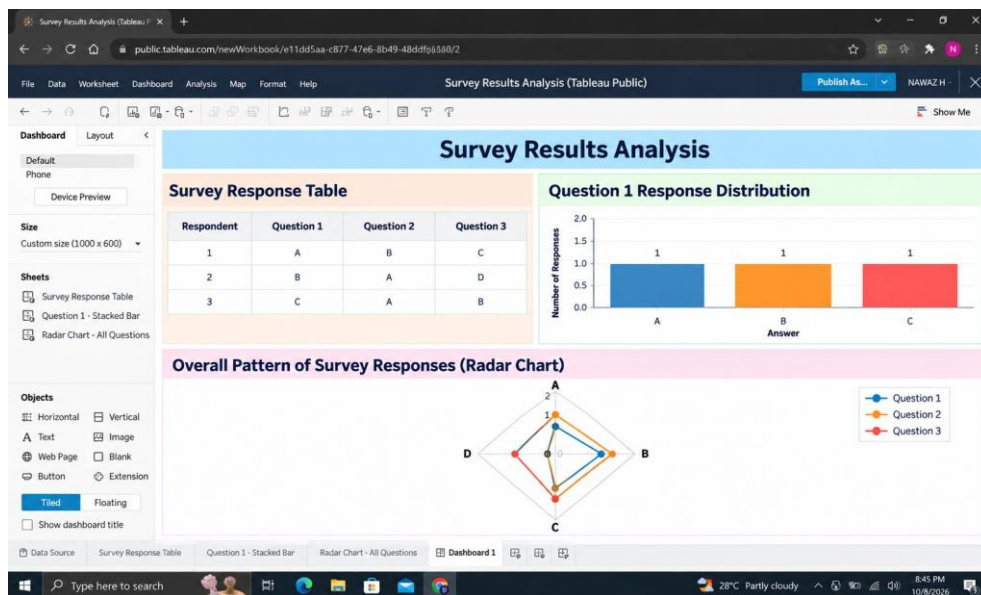
Console Output:

```
# Program 3: Survey Response Table
> 
> survey_data <- data.frame(
+   Respondent = c(1, 2, 3),
+   Question_1 = c("A", "B", "C"),
+   Question_2 = c("B", "A", "A"),
+   Question_3 = c("C", "D", "B")
+ )
> 
> print(survey_data)
  Respondent Question_1 Question_2 Question_3
1          1          A          B          C
2          2          B          A          D
3          3          C          A          B
```

Observation: The table displays the answers given by the three respondents to Questions 1, 2, and 3.

PROGRAM 4: Tableau Dashboard

Develop a Tableau dashboard combining the survey response table, stacked bar chart, and radar chart for interactive exploration.



Output 4: Tableau Survey Results Analysis Dashboard

Observation: The Tableau dashboard combines the survey response table, the Question 1 bar chart, and the radar chart to give an overall view of the survey results.

Procedure

1. Create the respondent, question, and answer vectors in R.
2. Count the responses for Question 1 using `table()`.
3. Use `barplot()` to generate and label the bar chart for Question 1.
4. Use `plot()`, `lines()`, `points()`, and `legend()` to build the radar chart for all questions.
5. Create and print the survey response table using `data.frame()`.
6. Enter the survey data into Tableau and create the required worksheets.
7. Combine the table, bar chart, and radar chart into a dashboard titled Survey Results Analysis.

Result

The survey response data was successfully visualized using a bar chart, radar chart, and table in R. A Tableau dashboard was also created by combining the survey response table, bar chart, and radar chart for interactive exploration.

Key R Functions Used

Function	Purpose
<code>c()</code>	Creates the respondent, question, and answer vectors.
<code>table()</code>	Counts the responses for each answer.
<code>barplot()</code>	Creates the bar chart for Question 1.
<code>plot()</code>	Sets up the plot area for the radar chart.
<code>lines()</code>	Draws the radar chart outlines.
<code>legend()</code>	Labels the questions on the radar chart.
<code>data.frame()</code>	Creates the survey response table.